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Transforming Text into Structured Knowledge: A Frame-Semantics and RDF-based Approach

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Abstract

A substantial portion of human knowledge is embedded in lengthy and heterogeneous texts—such as biographies, news articles, and encyclopedic entries—and transforming it into structured, queryable representations, particularly for Graph RAG applications, remains a major challenge. This process requires identifying events and semantic roles, linking entities to authoritative resources, resolving pronominal references, and ultimately mapping the knowledge into RDF triples. This research presents an end-to-end pipeline for generating knowledge graphs and RDF triples, which, by relying on frame-based representation (rather than shallow relation extraction), enables a more robust transformation from text to stable predicates. The pipeline is structured around three key components: semantic role and frame extraction with the Frame Semantic Transformer, entity linking with REL accompanied by a confidence threshold, and pronominal coreference resolution with LLaMA 2 via DeepInfra. This combination was chosen based on practical and domain-specific considerations: LLaMA 2 strikes a balance between accuracy and latency for short-to-medium texts under limited computational budgets, and, through the DeepInfra platform, it is accessible in a stable and reproducible manner. Moreover, large language models, due to their flexibility and ability to handle semantic ambiguities, are more suitable for this role than purely pre-trained models. The intermediate representation is then translated into predicates through predefined mappings, yielding RDF triples. The pipeline was evaluated on a dataset of one hundred Wikipedia biographies of writers. To ensure quality and unbiased, consistent evaluation of the triples, the “LLM-as-a-Judge” approach was adopted using DeepSeek V3, which assessed triples in terms of extractability from the text and confidence level. In total, 1,675 sentences were processed, 2,720 entities extracted, and 9,635 triples produced; among these, 4,743 triples (up to 50 per writer) were evaluated. The average confidence was 53.1% with a median of 60.0%; around 60% of the triples were above the median and deemed acceptable, while nearly 35% were evaluated as high quality. The concentration of results in the mid-range (0.4–0.7) indicates that strengthening extraction robustness and improving confidence calibration could enhance overall quality. Beyond producing biographical knowledge graphs, this pipeline has practical applications in supporting the retrieval layer of Graph RAG, enabling structured knowledge-based question answering, and reinforcing verifiability by linking responses to textual evidence. It thus provides a practical pathway for more precise utilization of long texts in knowledge-centric systems.

Keywords: *Knowledge Graph, RDF, Semantic Frames, Large Language Models, Entity Linking, Semantic Role Labeling, LLM-as-a-Judge*

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Chapter 1: Introduction

1.1 Problem Statement

In the real world, a substantial portion of human knowledge is accumulated in the form of lengthy and heterogeneous texts, ranging from biographies and news reports to encyclopedic entries and articles. Although search engines have facilitated access to these texts, converting raw content into structured representations suitable for inference, precise question answering, and integration into knowledge graphs remains a fundamental challenge.

The difficulty arises from the need to identify events within natural sentences along with their semantic roles, link entities to reference sources, resolve pronominal references and linguistic indications, and ultimately transform all this information into RDF-compatible triples. These are triples that can be queried on a knowledge graph and directly utilized in Retrieval-Augmented Generation (RAG) scenarios.

This process faces numerous practical limitations at web scale: the excessive length of inputs and the risk of request timeouts, semantic ambiguity of roles, polysemy of names, lack of sufficient context for resolving pronouns, differences in writing styles, and the heterogeneity of output formats.

1.2 Necessity and Application

In response to this challenge, this research designs an end-to-end pipeline that organizes the path from natural text to RDF triples and knowledge graphs in a reproducible manner. The core idea is that instead of relying directly on shallow relation extraction, semantic frames and their roles are extracted first. Then, these frames are translated into stable predicates and triple patterns. Along the way, pronominal references are attributed to the nearest valid entity using LLM-based coreference resolution.

Finally, a dataset of 100 Wikipedia texts related to top historical writers is extracted, the entire pipeline is applied to them, and after generating their triples, an "LLM-as-a-Judge" approach is used to ensure quality. Each triple is tagged based on its extractability from the text, confidence level, and a brief explanation, providing a precise picture of the entire chain's performance through quantitative metrics.

This pipeline, beyond constructing biographical and historical knowledge graphs and triples, has several direct and practical applications:

1. The generated triples with rich semantic structures enable more precise information retrieval in graph-based question-answering systems. These structures increase the capability to infer complex relationships and answer multi-step questions, laying the groundwork for more advanced intelligent systems.
2. Converting biographical texts into RDF triples enables the creation of queryable and extensible knowledge bases that can be used in various research and commercial projects. These knowledge bases allow for the development of integrated knowledge ecosystems through integration with existing systems and other knowledge bases.
3. Each triple is linked to its original text source, allowing for information tracking and validation. This feature is essential for increasing reliability and citability in intelligent systems, enabling evidence-based evaluation. Furthermore, the extracted triples facilitate the analysis of complex relationships between historical figures, literary works, and significant events, which is applicable in literary and historical studies.

1.3 Technical Challenges

This process faces several technical challenges, as outlined below:

1. Encyclopedic texts and biographies in Wikipedia are typically long and contain heterogeneous sections. For stable processing, there is a need for meaningful chunking while preserving sentence boundaries to prevent process interruption.
2. Identical names for people, places, or organizations require precise linking to reference sources, taking into account confidence thresholds and contextual evidence.
3. Coreference resolution and attributing pronouns to correct references are essential to prevent invalid nodes in the knowledge graph.
4. Translating semantic roles into stable graph predicates requires a comprehensive and extensible mapping dictionary to convert role labels into queryable relationships.
5. Instead of relying on mere string matching, human-like evaluation using an "LLM-as-a-Judge" allows for assessing extractability from the text and generating explanations.

1.4 Report Structure

The structure of this report is compiled step-by-step with the aim of reproducibility and is presented in a coherent narrative as follows:

1. **Introduction and Fundamental Concepts:** Precise problem statement and introduction of operational definitions in Natural Language Processing, Semantic Role Labeling, Entity Linking, Coreference Resolution, as well as the foundations of RDF, RAG, and Graph RAG, along with the idea of the LLM-as-a-Judge to formulate the theoretical framework and research vocabulary.
2. **Related Work:** A critical review of current research and tools in semantic role extraction, entity linking, mapping frames to graph relations, and Graph RAG approaches.
3. **Proposed Method:** Description of the pipeline architecture from collection and length control to frame/role extraction, linking and coreference resolution, transformation of the intermediate structure to RDF, and judge-based evaluation, clarifying constraints, assumptions, and data flows.

4. **Implementation:** Detailed description of input/output specifications for each step, data formats (extracted frames, identified entities, triples, and evaluation output structure), confidence thresholds, error handling and re-execution strategies, and optimization points such as text chunking, sequence length limits, and stability.
5. **Results and Future Directions:** Definition of metrics (extraction rate, weighted average confidence, and confidence distribution), successful examples of extracted triples along with judge model evaluation, and presentation of performance tables and charts separated by categories.

Chapter 2: Preliminaries

2.1 Introduction

In this chapter, the fundamental concepts necessary for a precise understanding of the subsequent discussions are explained in a coherent manner. The goal is to establish a common vocabulary and conceptual framework for the reader to follow the path "from raw text to structured representation." The concepts in this chapter are organized into five main clusters: Natural Language Processing, Semantic Concepts, Entity Linking, Semantic Web, and Machine Learning. Then, a set of advanced and specialized concepts that are particularly important for modern knowledge extraction and knowledge graph applications are presented.

2.2 Natural Language Processing (NLP) Concepts

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to analyze, understand, and generate human language. This field encompasses diverse tasks such as syntactic parsing, semantic analysis, summarization, translation, and knowledge extraction. The importance of NLP lies in its ability to create a bridge between raw textual data and computable representations.

2.2.1 Entity Extraction

Named Entity Recognition (NER) is a key task that involves identifying and categorizing segments of text such as names of people, places, dates, or works. The output of NER is often used as input for more advanced tasks like entity linking and knowledge graph construction. One of the main challenges in NER is managing polysemy and compound entities.

2.2.2 Coreference Resolution

Coreference resolution is the process of determining which entity in the text pronouns and referring expressions (such as "he", "they", "she") refer to. Correct coreference resolution is essential to prevent

the creation of incorrect nodes in the knowledge graph. Classical methods were rule-based, but today Large Language Models have shown impressive performance in this area.

2.3 Semantic Concepts

The goal of this section is to go beyond syntax and represent relationships and events at the semantic level. In linguistics, meaning is defined as the intellectual or informational content conveyed by sentences or phrases. This section analyzes roles and relationships that occur at the semantic level, examining not just the structural form of sentences but their semantic structure.

2.3.1 Semantic Roles

Semantic roles specify the role each word or phrase plays in an event (Agent, Patient, Instrument, Location, Time, etc.). This representation is independent of syntax and answers the question: "Who did what to whom, where, and when?"

2.3.2 Semantic Frames

A semantic frame is a conceptual structure that describes an event or situation along with its participants. For example, the `Commerce_buy` frame includes roles such as "Buyer", "Seller", "Goods", and "Money". Frames help organize knowledge scattered throughout the text.

2.3.3 FrameNet

FrameNet is a well-known database that documents an extensive collection of frames and semantic roles. This resource enables the training and evaluation of semantic role labeling models.

2.3.4 Semantic Role Labeling (SRL)

Semantic Role Labeling (SRL) is an automated process in which systems identify the appropriate frame for a sentence and assign roles to text components. This task is key for the transition from the syntactic level to the semantic level.

2.3.5 Fillmore's Frame Semantics

Fillmore's Frame Semantics theory is a key theory in linguistics that models the semantic roles of words within conceptual frames. According to this theory, the meaning of a word or phrase depends not only on the word itself but also on the conceptual frame in which the word is situated. These frames include a set of roles typically used in a specific event or situation.

2.4 Entity Linking Concepts

Entity Linking (EL) is the process of connecting extracted textual mentions to unique resources in knowledge bases. This is crucial for name disambiguation and knowledge integration.

2.4.1 Concept of Entity

In NLP, an entity refers to any name or object that can be independently identified and described in a text. These entities can include people, places, organizations, events, products, and abstract concepts. Entities are essentially considered references to specific objects or concepts in the real or conceptual world. For example, in the sentence "Albert Einstein was born in Bern, Switzerland," the name "Albert Einstein" is identified as a person entity and "Bern" as a location.

2.4.2 Entity Disambiguation

Entity Disambiguation is a subset of EL tasked with selecting the correct item among several identical names (e.g., "Lion" as an animal or a movie). This process is performed using sentence context, statistical features, and knowledge bases.

2.4.3 DBpedia and Wikipedia

Wikipedia is the primary textual source for identifying entities, and projects like DBpedia have extracted structured data from Wikipedia. Connecting textual entities to Wikipedia articles or DBpedia URIs is one of the most common linking methods.

2.4.4 URI

A Uniform Resource Identifier (URI) is a standard string used to identify resources on the Semantic Web. URIs ensure that entities are referenced uniquely and persistently.

2.5 Semantic Web Concepts

The Semantic Web is a vision that makes data understandable and inferable for machines. Key standards include RDF, SPARQL, and serialization formats like Turtle.

2.5.1 RDF and Triples

The Resource Description Framework (RDF) is a framework for representing knowledge as Subject-Predicate-Object triples. These triples are considered the smallest units of knowledge in the Semantic Web.

2.5.2 Knowledge Graph

A Knowledge Graph is a structured representation of information where nodes represent entities (such as people, organizations, places, works, or abstract concepts) and edges represent the relationships between them. A key feature of a knowledge graph is the combination of scattered textual information into a unified network, enabling inference, multi-hop querying, and the discovery of hidden connections. Knowledge graphs are usually built on standards like RDF, making data formally processable. One major advantage is the ability to integrate data from heterogeneous sources; for example, combining a writer's biographical information from Wikipedia with bibliographic data from other databases.

2.6 Machine Learning and Language Model Concepts

Machine Learning, and specifically Deep Learning, has played a significant role in the advancement of NLP. The required concepts in this field are briefly described below.

2.6.1 Language Models

Language Models (LMs) are systems that learn the probability distribution of sequences of words. Large Language Models (LLMs) like GPT have taken text understanding and generation to a new level.

2.6.2 Transformer Architecture

The Transformer architecture revolutionized NLP. Relying on the Attention mechanism, this architecture is capable of modeling long-range dependencies in text. Its main components include Feed-

Forward layers, Self-Attention, and Positional Encoding. Almost all modern models (BERT, GPT, T5) are built on this architecture.

2.6.3 LLM-as-a-Judge

"LLM-as-a-Judge" is a novel approach in evaluating data extracted from NLP systems. In this method, a Large Language Model acts as a judge, evaluating system outputs (like RDF triples) along with textual evidence. The model assesses the quality of extracted data using criteria such as confidence level and a brief reason. This method allows for human-like judgment alongside statistical evaluations. In this project, LLMs are used to evaluate the "extractability" of each triple, and only triples extracted with a confidence level above a certain threshold are accepted as valid results.

2.6.4 Confidence Score

The Confidence Score is a metric indicating the model's certainty in a prediction or decision. In tasks like EL and SRL, only outputs with a score higher than a threshold are accepted as valid. In this project's evaluation section, this metric equals the confidence level provided by the Judge model regarding the extractability of each triple.

2.7 Chapter Conclusion

In this chapter, the fundamental concepts and theoretical framework required to understand the path from raw text to structured text were introduced. First, by reviewing NLP and its key tasks such as entity extraction, coreference resolution, and semantic role labeling, it was shown how text can be represented at a richer semantic level. Then, semantic concepts such as Fillmore's roles and frames and resources like FrameNet were introduced as a basis for the transition from the syntactic to the semantic level. Next, the importance of linking entities and disambiguating them using resources like Wikipedia and DBpedia was stated. Furthermore, Semantic Web concepts, standards like RDF, and the structure of knowledge graphs were explained, providing a platform for storing and inferring knowledge. Finally, the role of Machine Learning, especially Large Language Models, in advancing the aforementioned tasks and the idea of using the "LLM-as-a-Judge" as a novel evaluation approach were explained. These topics established the necessary theoretical framework for the current research and created a common vocabulary that will serve as the basis in subsequent chapters. In the next chapter, related work will be reviewed to clarify the position and innovation of the proposed approach compared to the background by reviewing current research and tools in the mentioned fields.

Chapter 3: Related Work

3.1 Introduction

In the previous chapter, the foundational concepts of Natural Language Processing, Semantics, Semantic Web, and Machine Learning were introduced. In this chapter, related works that are directly relevant to the path of this research—namely, the transition from raw text to structured representations in the form of RDF triples and their utilization in Graph RAG—are reviewed.

The main focus is on four axes:

1. **Entity Linking and Disambiguation:** Research and tools that play a fundamental role in connecting textual data to knowledge bases.
2. **Frame and Semantic Role Extraction:** Their application in enriching knowledge graphs.
3. **Automated Knowledge Graph Construction:** Novel approaches for building graphs from unstructured text, combining statistical and symbolic methods to enhance representation quality.
4. **Integrating Knowledge Graphs with LLMs (Graph RAG):** Approaches enabling multi-step reasoning and documented question answering.

The goal of this chapter is to provide a comprehensive overview of recent advancements in these fields and position the current project within these efforts, specifically emphasizing methods that have effectively bridged the semantic layer and the capabilities of interactive language models.

3.2 Entity Linking

The SPEL method defines entity linking as a structured prediction problem and, using a two-stage optimization, has achieved high accuracy on standard datasets like AIDA. The importance of this work lies in its lightweight nature and its ability to simultaneously utilize local and contextual cues, making it suitable for long and complex texts. [\[11\]](#)

The OneNet model proposed a different path, demonstrating that Large Language Models (LLMs) can perform entity linking solely through in-context learning without fine-tuning. The design of dual text and history views in this model reduces errors caused by surface similarities and increases efficiency across different domains. [8]

Furthermore, research such as EntGPT has shown that by designing multi-step queries for LLMs, the entity linking process can be performed with quality comparable to supervised methods. [3] Despite these advancements, comprehensive reviews in the biomedical domain have indicated that name disambiguation in specialized domains remains challenging and requires context-aware methods combined with coreference resolution. [7]

3.3 Frame Extraction and Frame-Based Knowledge Graphs

Extracting semantic frames provides an efficient solution for representing complex, multi-role relationships between entities. The FRED system was one of the first efforts in this field, demonstrating the possibility of generating queryable knowledge graphs from free text by combining semantic analysis with standard formats like RDF/OWL. The significance of FRED was in establishing a bridge between Semantic Role Labeling (SRL) and Semantic Web representations. [5]

Subsequently, projects like PRIVAFRAME showed that using frames for specific domains, such as data privacy, can improve the accuracy of detecting sensitive information compared to purely neural models, while also offering greater transparency and explainability. [4] Additionally, initiatives like FrameBase [10] and TakeFive [1] have attempted to facilitate the direct transformation from text to standard triples by creating mappings between semantic roles and graph structures. These collective studies demonstrate that frames not only provide a deeper understanding of events but are also key tools for building precise and reliable knowledge graphs.

3.4 Automated Knowledge Graph Construction from Text

In the domain of automated knowledge graph construction, the EDC (Extract, Define, Canonicalize) framework presented a three-stage approach that produces accurate outputs compatible with existing schemas without the need for supervised training. [14] The importance of this method lies in mitigating the problem of repetition and inconsistency in triples generated by language models, enabling its use on long and heterogeneous texts.

Alongside this approach, neuro-symbolic research has shown that combining linguistic descriptions generated by language models with formal representations like AMR and RDF/OWL, and using a feedback loop, can produce multi-layered, rich graphs that possess higher reasoning capabilities in addition to greater accuracy. [2] These works are of particular importance to the current research as they demonstrate that combining statistical and symbolic mechanisms is the best path to reduce common errors of language models in knowledge graph generation.

3.5 Graph RAG: Integrating Knowledge Graphs with Large Language Models

One of the emerging and highly prominent trends in recent research is the integration of knowledge graphs with LLMs in the form of Graph RAG. The GRAG method, by introducing a divide-and-conquer approach for retrieving relevant subgraphs and simultaneously injecting textual and structural representations, has achieved performance surpassing textual RAG in multi-step questions. [6] The significance of this work is enabling inference based on graph paths, going beyond mere document retrieval.

At the application level, the Microsoft GraphRAG project has demonstrated that combining graphs extracted from organizational corpora with hierarchical clustering and node summarization can provide

documented and traceable answers, thereby reducing the problem of hallucination and mistrust in text generation.[9] Additionally, methods like KG-Adapter [12] and KG-CoT [15] have offered different paths for integrating graph knowledge into language models; the former by adding lightweight adapter layers to decoder-only models, and the latter by guiding the chain of thought through graph traversal. Alongside these, the classic QA-GNN work has highlighted the importance of Graph Neural Networks (GNNs) combined with language models for question answering.[13] Collectively, this body of research emphasizes that the future of RAG lies in utilizing graph structures, and the current research, by producing semantically-grounded triples and connecting them to Graph RAG, aligns precisely with one of today's hottest scientific trends.

3.6 Chapter Conclusion

In this chapter, a collection of research and related approaches that have shaped the path from raw text to structured representations and knowledge graphs was reviewed. First, modern methods demonstrated how leveraging local and contextual cues or the capabilities of LLMs can achieve better accuracy and coverage in connecting entities to reference databases. Second, the mapping of semantic frames to Semantic Web structures was introduced as a foundation for a more precise representation of events and multiple roles. Third, studies on constructing knowledge graphs from text, particularly neuro-symbolic approaches, indicated that combining statistical and symbolic representations is the optimal path for reducing error and increasing reasoning power. Finally, the review of Graph RAG approaches showed that combining graph knowledge with LLMs is a leading research trend capable of resolving the limitations of textual RAG.

This review clarified the position of the current research: relying on semantic frames to enrich triples, linking entities for knowledge integration, and utilizing an "LLM-as-a-Judge" for evaluation all address the gaps and needs highlighted by related works.

In the next chapter, the **"Proposed Method"** will be introduced; the pipeline architecture, its components, and the logic behind tool selection will be described in detail to demonstrate how these gaps can be filled to provide a practical and reproducible framework for knowledge extraction.

Chapter 4: Proposed Method

4.1 Introduction

In this chapter, an integrated approach for extracting knowledge from text and transforming it into RDF triples is described. This approach relies on the systematic integration of three key components: event-centric representation via semantic role labeling, precise linking of nominal mentions to reference entities, and final judging and refinement using Large Language Models (LLMs).

4.2 Core Idea of the Proposed Method

The general workflow is as follows: Raw text is first converted into an intermediate Frame-Role representation to make the structure of events and participants (Agent, Object, Time, Place, etc.) explicit. Then, named mentions are mapped to knowledge-based identifiers to elevate references from a superficial lexical level to a unique semantic level. Subsequently, through coreference resolution, links between pronouns and ambiguous references are established with the nearest valid entity. Finally, this intermediate representation is transformed into standard RDF triples based on semantic mappings from frames and roles to predicates.

To ensure stability and quality, one hundred biographies of top writers were fetched from Wikipedia. After length control and sentence-boundary chunking, they were processed through the pipeline, incorporating ambiguity reduction strategies throughout. The outputs are then assessed by a human-like evaluation protocol based on an LLM, measuring "extractability" and "confidence". The strength of this design lies in ensuring both semantic accuracy and integration capability into knowledge graphs, while the separation of extraction, linking, and judging layers allows for gradual development and easy component replacement.

4.3 Technologies Used

This section outlines the technologies used in the project implementation, along with the rationale and advantages of each.

4.3.1 Frame Semantic Transformer Model

The Frame Semantic Transformer is a network based on the transformer architecture, specifically trained for the task of Semantic Role Labeling (SRL). By utilizing deep textual representations, this model is capable of extracting semantic frames such as Text_creation, Being_born, and Categorization, along with their associated roles like Author, Text, Time, and Place from sentences. The main advantage of this model over classical approaches is its better coverage of diverse textual domains and higher accuracy in identifying peripheral roles. Furthermore, the use of the transformer encoder structure enhances the model's generalization power and applicability to new texts.

4.3.2 REL Model

The REL (Radboud Entity Linker) model is a lightweight system for linking entities to knowledge bases (e.g., Wikipedia). This system first extracts all candidate mentions (such as names of people, places, and organizations) from the text and then suggests multiple candidate entities based on local and contextual cues. In the next step, these candidates are ranked, and a confidence score is calculated for each link. This score allows for the application of a threshold to filter out low-confidence items, keeping only valid and accurate links. The model output includes the mention name, corresponding entity, Wikipedia identifier, and confidence value. In addition to high accuracy in disambiguation, REL produces refined and reliable outputs due to its lightweight nature and thresholding capability, making it easily integrable into NLP pipelines.

4.3.3 LLaMA 2 Model for Coreference Resolution

Large Language Models are a better choice for tasks like entity identification and resolving semantic ambiguities due to their high flexibility, better adaptability to specific data, and strong performance in processing complex language compared to purely pre-trained models. LLaMA 2 was selected for its high reasoning capability and understanding of long-range dependencies in complex texts, specifically for coreference resolution and pronoun disambiguation. This model can identify the precise reference of pronouns and extract related entities from the text with high accuracy. Moreover, LLaMA 2's scalable architecture provides suitable processing power for handling high data volumes while maintaining efficiency at web scale. Since LLaMA 2 is called via the DeepInfra platform, access is stable and reproducible across different computing environments. These features make LLaMA 2 an optimal choice for precise coreference resolution in long and complex texts.

4.3.4 DeepSeek V3 Model for Triple Accuracy Evaluation

The DeepSeek V3 model was selected as a powerful tool for semantic judging and textual inference to evaluate the accuracy of extracted triples. This model is highly capable of assessing the confidence level of each triple and clearly articulating the logical reason for its judgment, which is essential for ensuring data correctness and quality. Using DeepSeek V3 as an "LLM-as-a-Judge" ensures that evaluations remain immune to the self-confirmation biases of extraction models, improving evaluation accuracy. Furthermore, with its ability to accept implicit inferences from text, this model enables more precise and interpretable evaluations in complex conditions. Using a judge language model is particularly preferable here due to the lack of labeled datasets and the need for flexible, accurate evaluation in unstructured and unsupervised settings (like complex semantic relationships). Models like DeepSeek V3 can automatically and with precise text understanding perform evaluations without reliance on labeled data, providing better results in assessing triple quality.

4.4 Conceptual Pipeline

Conceptually, the text's event structure is first extracted using the Frame Semantic Transformer. These models, trained on frame-centric data, are capable of identifying action units and assigning core and

peripheral roles, demonstrating resilience to syntactic shifts and writing styles. Their main advantage is creating a rich intermediate representation map-able to semantic relationships.

Next, REL connects names and nominal phrases to stable identifiers in reference databases (Wikidata/Wikipedia). REL balances accuracy and coverage by combining local and textual evidence, and its confidence scoring allows for threshold-based filtering. This eliminates low-confidence links and protects the final graph from reference errors.

For coreference resolution, DeepInfra is utilized as a capable LLM provider to determine the reference of pronouns and ambiguous mentions based on textual reasoning and semantic consistency.

The output (roles and linked entities) is converted into RDF using semantic mappings (from frames/roles to predicates) and domain ontologies. Finally, an LLM-based evaluation framework tags the triples, reporting metrics such as accuracy, extraction rate, weighted average confidence, and confidence distribution. This report enables both error analysis and approach comparison.

To this end, the combination of DeepSeek V3 (for evaluation), Frame Semantic Transformer (for event structuring), REL (for precise linking), and DeepInfra (for coreference) creates a balanced mix of accuracy, coverage, structural transparency, and operational efficiency, making it reliable for structuring specialized texts and integrating them into knowledge graphs.

4.5 Chapter Conclusion

In this chapter, the proposed research method was introduced as an integrated pipeline. First, the core idea of the approach—transitioning from raw text to frame-role representation, linking entities, resolving coreference, and finally generating RDF triples—was described. Then, key technologies including Frame Semantic Transformer for SRL, REL for entity linking, LLaMA 2 for coreference resolution, and DeepSeek V3 for triple judgment were introduced, explaining the advantages of each. The conceptual pipeline was also presented, clearly showing data flow and component interaction. This approach, relying on semantic representation, precise linking, and human-like evaluation, provides a balanced framework of accuracy, coverage, and inferential capability suitable for constructing knowledge graphs from long texts. In the next chapter, "**Implementation**" details will be provided, describing the practical execution of each component and data formats.

Chapter 5: Implementation

5.1 Implementation Overview and Goals

In this chapter, the implementation details of the knowledge extraction pipeline—from text-based input to RDF triple generation—are presented in a practical and reproducible manner. The goal is to demonstrate the precise path from raw data to structured knowledge. To this end, the mechanisms for data preparation, semantic role extraction, entity linking, coreference resolution, semantic mapping of frames to predicates, and final transformation to RDF are described. Finally, the evaluation protocol, result recording, error handling mechanisms, re-execution strategies, and computational optimization notes are provided.

5.2 Environment Preparation and Dependencies

To ensure stable and reproducible execution, the computing environment is configured within a virtual environment, and library versions are locked. The core packages include Natural Language Processing tools, deep learning frameworks, and knowledge graph libraries. This layer allows for independent execution of each step and flexible component replacement. The required libraries include `matplotlib`, `beautifulsoup`, `requests`, `frame-semantic-transformer`, `spacy`, `nlk`, `rdflib`, and `graphviz`. Installation is handled by a script that installs dependencies for each module in its dedicated environment.

5.3 Semantic Role Extraction

Frame Semantic Transformer models are used to extract the event structure of the text. As the starting point of processing, this module splits the text into independent sentences and extracts a set of semantic frames and associated role elements (such as Place, Time, Text, Author) for each sentence.

5.3.1 SRL Module Implementation The SRL module is implemented relying on the `frame_semantic_transformer` library. It first splits the input text into sentences using the NLTK tool. Then, each sentence is sent to the Frame Semantic Transformer model, and frames are extracted along with role elements. The output is organized in JSON format, storing identified frames, roles, and corresponding text values for each sentence. The design allows for using both small and base model sizes to balance speed and accuracy.

5.3.2 SRL Module Output Structure The SRL module maps the textual input to an array of sentences. For each item, the `text` key holds the full sentence text, and the `frames` key provides a list of extracted semantic frames. Each frame is described by `name` and an array of role elements. In each element, the

Sample Output:

```
{
  "sentences": [
    {
      "text": "Born in Moscow in 1821, Dostoevsky was introduced to...",
      "frames": [
        {
          "name": "Being_born",
          "elements": [
            { "name": "Place", "text": "in Moscow" },
            { "name": "Time", "text": "in 1821" },
            { "name": "Child", "text": "Dostoevsky" }
          ]
        }
      ]
    }
  ]
}
```

`text/name` pair records the semantic role name (e.g., Time, Place) and the corresponding text segment.

5.4 Entity Linking

5.4.1 REL Module Implementation

The REL module is used to link entities to knowledge bases like Wikipedia and disambiguate names. It first divides the text into smaller chunks (Chunking) to make extraction from long texts more efficient. Each chunk is sent to the REL API, where entities are tagged and a confidence score is calculated. A default confidence threshold of 0.5 is applied to filter out low-confidence entities. Each identified entity is assigned a unique identifier (URI) from Wikipedia.

5.4.2 REL Module Output Structure

The output is a JSON object with an `entities` key, containing an array of "mention-entity" objects. Each object includes `mention` (textual reference), `entity` (Wikipedia page title), `uri` (full link), `start/end` (character positions), and `confidence`.

Sample Output:

```
{
  "entities": [
    {
      "mention": "Dostoevsky",
      "entity": "Fyodor_Dostoevsky",
      "uri": "http://en.wikipedia.org/wiki/Fyodor_Dostoevsky",
      "confidence": 0.9239
    },
    ...
  ]
}
```

5.5 RDF Triple Generation

The RDF module forms the link between linguistic processing and knowledge graph construction. It generates Subject-Predicate-Object triples based on frame analysis and entity linking.

5.5.1 RDF Module Implementation

The implementation follows two complementary paths: a Dictionary-based path for frequent frames and a Generic path for others.

1. **Index Construction:** A unique index of identified entities is built to map synonymous references to a single ID.
2. **Coreference Resolution:** Performed via DeepInfra to connect pronouns to valid references before assignment.
3. **Dictionary Path:** For ~50 frequent frames, roles are directly mapped to specific predicates (e.g., `FrameName:has_<Role>`).
4. **Generic Path:** For other frames, a subject is determined (prioritizing person-centric roles), and generic predicates are created for each role.

Table 5-1: Sample Mapping Rules

Semantic Frame	Input Roles	Generated RDF Triple
Being born	Child, Time, Place	<code>Child Being_born:has_time Time</code> <code>Child Being_born:has_location Place</code>
Death	Deceased, Time, Place	<code>Deceased Death:has_time Time</code> <code>Deceased Death:has_location Place</code>
Travel	Traveler, Destination	<code>Traveler Travel:has_destination Destination</code> <code>Traveler Travel:has_time Time</code>

5.5.2 Coreference Resolution with LLM

To disambiguate pronouns, a hybrid solution using **LLaMA 2** on DeepInfra is employed. The module sends the ambiguous pronoun along with the sentence and paragraph to the model to identify the precise reference.

5.5.3 RDF Module Output Example

Output is stored as linearized triples. Subjects and objects are URIs if successfully linked; otherwise, they appear as text strings.

Example: `http://en.wikipedia.org/wiki/Fyodor_Dostoevsky Being_born:has_location http://en.wikipedia.org/wiki/Moscow`

5.6 Pipeline Execution on Authors' Biography Dataset

A dataset of 100 top writers was collected from Wikipedia to evaluate the pipeline.

5.6.1 Text Collection and Refinement

Texts were fetched using a Wikipedia scraper script. Irrelevant sections (footnotes, tables) were removed, and text length was limited to a maximum of 2000 characters, truncated at sentence boundaries to preserve semantic integrity .

5.6.2 Pipeline Execution

The texts entered the main pipeline: SRL → REL → RDF conversion (with coreference resolution) → Visualization → Evaluation. This process was managed by a BatchPipeline module, supporting both parallel and sequential processing.

5.6.3 Evaluation with DeepSeek V3

The TripleEvaluator module sends RDF triples to the DeepSeek V3 model. For each triple, the model reports:

- **Extractable:** (Boolean) Whether it is justifiable from the text.
- **Confidence:** (0-1) Reliability score.
- **Reason:** A brief explanation.

5.6.4 Evaluation Output Structure

```
{  
  "triple": "...",  
  "extractable": true,  
  "confidence": 0.9,  
  "reason": "The paragraph explicitly states 'Born in Moscow in 1821'..."  
}
```

5.6.5 Storing Results

Results are saved as JSON files containing triple details, validity percentages, and confidence distributions. Finally, enriched CSV files are generated for RAG applications.

5.7 Knowledge Graph Construction

The final step involves visualizing the extracted data as a graph where nodes represent entities and edges represent semantic relationships. Tools like rdflib and Graphviz are used to visualize key relationships (e.g., birth, death, works).

5.8 Chapter Conclusion

In this chapter, the practical implementation of the proposed pipeline was detailed. Environment setup, core modules (SRL, REL, RDF, Coreference), output structures, and the evaluation process with the

"LLM-as-a-Judge" were described. Finally, the construction of knowledge graphs was explained. In the next chapter, the quantitative and qualitative results of this implementation will be analyzed.

Chapter 6: Results

6.1 Introduction

The objective of this chapter is to present and analyze the outputs obtained from the implementation of the knowledge extraction pipeline. As described in previous chapters, the project input consisted of a collection of biographical texts derived from Wikipedia, which, after collection and preprocessing, underwent semantic frame extraction, entity linking, coreference resolution, and finally transformation into RDF triples. In this chapter, the results are examined both quantitatively and qualitatively. On one hand, general metrics such as the number of processed sentences, extracted entities, identified frames, and generated triples are reported; on the other hand, the quality of outputs is analyzed based on the "LLM-as-a-Judge" evaluation in terms of accuracy and confidence levels. Additionally, examples of textual and visual outputs from various sections, along with an analysis of some results, are provided to further familiarize the reader with the project's goals and outcomes.

6.2 Selection of Experimental Data Domain

One of the primary criteria for selecting the experimental data domain is the linguistic and thematic diversity of the texts. Biographies of writers on Wikipedia are considered a suitable option from this perspective because: (1) They typically contain long and descriptive sentences, providing a suitable context for evaluating semantic role labeling and extraction. (2) They cover various types of entities, including people, works, dates, and places. (3) Relationships between entities in these texts are sufficiently rich; for example, "X published book Y in year Z" can be converted into an RDF triple. (4) They follow a semi-formal and uniform style, facilitating text standardization in NLP pipelines.

Based on this, selecting the domain of writers simultaneously provides two key advantages: on one hand, the data possesses sufficient diversity and richness to test the pipeline's capabilities, and on the other hand, due to consistency and accessibility on Wikipedia, it allows other researchers to reproduce the results.

6.3 General Processing Statistics

To evaluate the pipeline, a collection comprising 100 writers was processed. Table 6-1 shows a general overview of the quantitative metrics extracted.

Table 6-1: General Processing Statistics of the Pipeline

Indicator	Value
Processed Authors	100
Average Length of Biography Texts	~2,000 characters
Total Sentences	1,675
Total Entities	2,720
Total Semantic Frames	5,556
Total Generated Triples	9,635
Evaluated Triples	4,743 (Max 50 per author)

6.4 Analysis of Triples Confidence Distribution

To qualitatively evaluate the outputs, the distribution of confidence values for the evaluated triples is displayed in Figure 6-1. In this chart, the horizontal axis represents confidence levels, and the vertical axis indicates the number of triples evaluated with a confidence level equal to or higher than the examined value. The average confidence value is calculated to be approximately **0.53**, with a median close to **0.60**. These results indicate that the confidence level for a large portion of the outputs lies in the mid-range, and the system has performed relatively stably in extracting valid relationships.

Based on the analysis, about **60%** of the triples have a confidence score equal to or higher than 0.5, allowing them to be considered acceptable outputs. Furthermore, nearly **35%** of the triples surpassed the 0.8 threshold, indicating a higher level of accuracy and reliability. However, the concentration of a significant portion of triples in the mid-range (0.4 to 0.7) suggests that while high-quality outputs exist, a portion of the extracted relationships requires improvement in robustness and increased confidence levels. Overall, the observed distribution suggests that the pipeline has successfully converted a significant share of textual knowledge into valid triples, while leaving clear room for improvement in interpreting more complex relationships and highly ambiguous sentences.

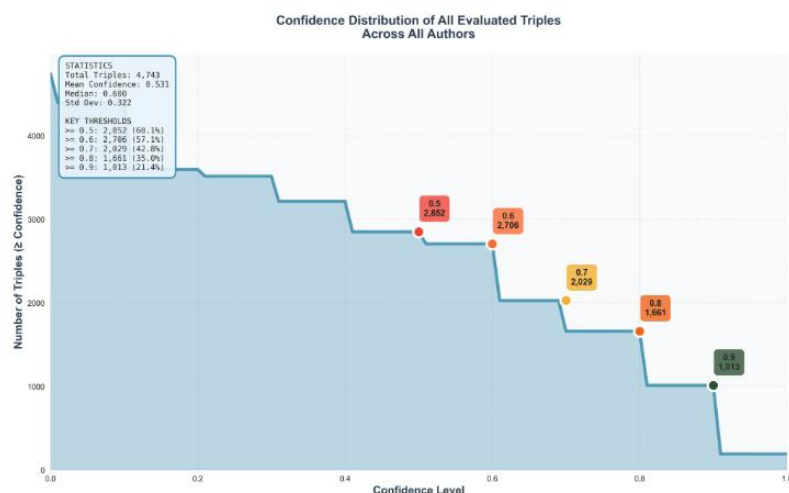


Figure 6-1: Confidence distribution of all evaluated triples across all authors

6.5 Successful Examples of Evaluated Triples

In the following, examples of evaluated triples for 5 authors are presented along with the results obtained from the judge model.

6.5.1 Leo Tolstoy

Below are examples of successful extracted triples related to Leo Tolstoy along with the judge model evaluation result.

```
{
  "triple": "http://en.wikipedia.org/wiki/Leo_Tolstoy Being_born:has_time 1828",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text explicitly states that Tolstoy was born in 1828, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Leo_Tolstoy Death:has_time 1910",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly states that Tolstoy died in 1910, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Leo_Tolstoy Categorization:has_category one of the greatest and most influential authors of all time",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text explicitly states that Tolstoy was one of the greatest and most influential authors of all time, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Leo_Tolstoy Marriage:has_spouse http://en.wikipedia.org/wiki/Sophia_Tolstaya",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text mentions Tolstoy married Sophia Behrs, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Leo_Tolstoy Belief:has_belief Christian anarchism",
  "extractable": true,
  "confidence": 0.85,
  "reason": "The text context suggests Tolstoy held Christian anarchist beliefs, providing a clear semantic connection."
}
```

6.5.2 Alexander Pushkin

Below are examples of successful extracted triples related to Alexander Pushkin along with the judge model evaluation result.

```
{
  "triple": "http://en.wikipedia.org/wiki/Alexander_Pushkin Being_born:has_location http://en.wikipedia.org/wiki/Moscow",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly states that Pushkin was born in Moscow, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Alexander_Pushkin Being_born:has_time 1799",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text context suggests Pushkin was born in the late 18th century, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Alexander_Pushkin Text_creation:has_text his controversial poem Ode to Liberty",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly mentions that Pushkin wrote his controversial poem 'Ode to Liberty', providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Alexander_Pushkin Categorization:has_category be the greatest Russian poet, as well as the founder of modern Russian literature",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text explicitly states that Pushkin is considered the greatest Russian poet and founder of modern Russian literature, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Alexander_Pushkin Death:has_time 1837",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text context suggests Pushkin died in the 1830s, providing a clear semantic connection for the 'Death' event having the time 1837."
}
```

6.5.3 Fyodor Dostoevsky

Below are examples of successful extracted triples related to Fyodor Dostoevsky along with the judge model evaluation result.

```
{
  "triple": "http://en.wikipedia.org/wiki/Fyodor_Dostoevsky Being_born:has_location http://en.wikipedia.org/wiki/Moscow",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly states that Dostoevsky was born in Moscow in 1821, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Fyodor_Dostoevsky Publishing:has_work His Notes from Underground, a novella",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly states that Dostoevsky published 'Notes from Underground, a novella published in 1864', providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Fyodor_Dostoevsky Text_creation:has_time mid-1840s",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text explicitly states that in the mid-1840s, Dostoevsky wrote his first novel, Poor Folk, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Fyodor_Dostoevsky Being_employed:has_position as an engineer",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text explicitly states that after graduating, Dostoevsky worked as an engineer, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Fyodor_Dostoevsky Sentencing:has_theme death",
  "extractable": true,
  "confidence": 0.8,
  "reason": "The text explicitly states that Dostoevsky was sentenced to death, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Fyodor_Dostoevsky Text:has_text Crime and Punishment (1866), The Idiot (1869), Demons (1872), The Adolescent (1875) and The Brothers Karamazov (1880)",
  "extractable": true,
  "confidence": 0.8,
  "reason": "The text explicitly mentions Crime and Punishment (1866), The Idiot (1869), Demons (1872), The Adolescent (1875) and The Brothers Karamazov (1880) as Dostoevsky's most acclaimed novels, providing a clear semantic connection."
}
```

6.5.4 Gabriel García Márquez

Below are examples of successful extracted triples related to Gabriel García Márquez along with the judge model evaluation result.

```
{
  "triple": "http://en.wikipedia.org/wiki/Nobel_Prize_in_Literature Receiving:has_recipient http://en.wikipedia.org/wiki/Gabriel_García_Márquez",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly states that García Márquez was awarded the 1982 Nobel Prize in Literature, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Neustadt_International_Prize_for_Literature Receiving:has_time 1972",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The paragraph clearly states 'he was awarded the 1972 Neustadt International Prize for Literature', providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Gabriel_García_Márquez Activity_start:has_role as a journalist",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly states that García Márquez started as a journalist, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Gabriel_García_Márquez Text_creation:has_text many ac
claimed non-fiction works and short stories",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text explicitly states that García Márquez wrote many acclaimed non-fiction wo
rks and short stories, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/Gabriel_García_Márquez Fame:has_reason his novels, su
ch as No One Writes to the Colonel (1961), One Hundred Years of Solitude (1967), which has sol
d over fifty million copies worldwide, Chronicle of a Death Foretold (1981), and Love in the Ti
me of Cholera (1985)",
  "extractable": true,
  "confidence": 0.8,
  "reason": "The text directly states that García Márquez is famous and best known for these sp
ecific novels, providing a clear semantic connection."
}
```

6.5.5 William Shakespeare

Below are examples of successful extracted triples related to William Shakespeare along with the judge model evaluation result.

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Being_born:has_time 1564",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text explicitly states that Shakespeare was born in 1564, providing a clear se
mantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Death:has_time 1616",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text explicitly states that Shakespeare died in 1616, providing a clear semant
ic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Text:has_text https://en.wikipedi
a.org/wiki/Romeo_and_Juliet",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text mentions 'Romeo and Juliet' as one of Shakespeare's famous plays, providi
ng a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Speaking:has_audience Theatre Audiences",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text mentions Shakespeare spoke to theatre audiences through his plays, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Speaking:has_time 1590s-1610s",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text context suggests Shakespeare's speaking to theatre audiences occurred during the 1590s-1610s, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Meeting:has_participant Theatre Company",
  "extractable": true,
  "confidence": 1.0,
  "reason": "The text mentions Shakespeare met with his theatre company, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Meeting:has_time 1590s-1610s",
  "extractable": true,
  "confidence": 0.8,
  "reason": "The text context suggests Shakespeare's meetings with his theatre company occurred during the 1590s-1610s, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Belief:has_belief Renaissance Humanism",
  "extractable": true,
  "confidence": 0.9,
  "reason": "The text context suggests Shakespeare held Renaissance humanistic beliefs, providing a clear semantic connection."
}
```

```
{
  "triple": "http://en.wikipedia.org/wiki/William_Shakespeare Belief:has_time 1590s-1610s",
  "extractable": true,
  "confidence": 0.8,
  "reason": "The text context suggests Shakespeare's Renaissance humanistic beliefs were prominent during the 1590s-1610s, providing a clear semantic connection."
}
```

6.6 Knowledge Graph Visualization

In this section, after extracting and generating triples, the knowledge graphs for each writer were visualized. These graphs reveal various semantic relationships such as times, places, people, and concepts present in the biographies and works of the authors.

Sample of generated knowledge graphs:

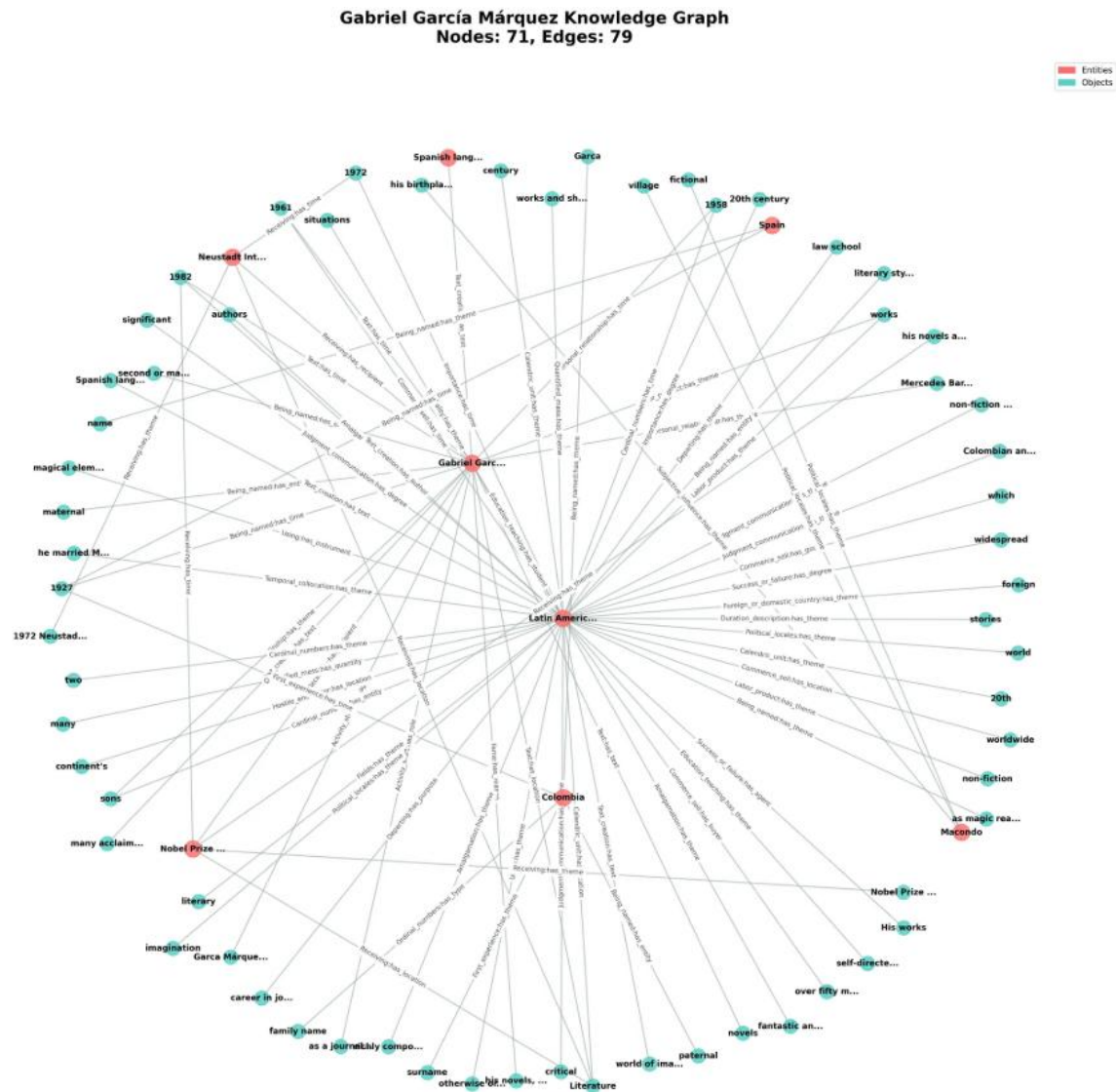


Figure 6-2: Gabriel García Márquez Knowledge Graph; receipt of literary awards, journalism activity, and his prominent novels are observable.

(More sample of graph visualizations are available in the original document)

6.7 Future Work

Given the results obtained from the implementation and evaluation of the knowledge extraction pipeline, there are multiple areas for improvement and further development that can enhance its performance and expand its application domain.

6.7.1 Increasing Extraction Accuracy

Results show that while 35% of triples are high quality, the high concentration in the mid-confidence range (0.4-0.7) indicates a need for mechanisms to strengthen extraction robustness. Prioritizing advanced deep learning methods to reduce deviations from high-confidence intervals and improving algorithms to identify complex relationships is essential.

6.7.2 Improving Semantic Role Labeling

One of the main areas for improving extraction quality is the use of more advanced Semantic Role Labeling models. These models, by utilizing more training data, especially data specific to different languages, can significantly increase frame detection accuracy. Using larger LLMs and improving frame extraction techniques can lead to better identification and separation of semantic frames.

6.7.3 Improving Entity Linking

The accuracy of entity linking is a main pillar of result quality. Leveraging richer knowledge sources, particularly large databases like Wikidata and DBpedia, and employing advanced matching algorithms can enhance linking accuracy. Additionally, using deep learning-based methods for matching entities and identifying unknown entities can help refine final extractions.

6.7.4 Expanding Application Domain

One important step for expanding the pipeline's application is supporting multiple languages. This is particularly important for international applications and can create multilingual capabilities in the system. Also, testing the pipeline in diverse domains such as medicine, law, and technical sciences can help improve the system's accuracy in processing varied text types (news, scientific, literary).

6.7.5 Improving Evaluation and Validation

Using comparative evaluation with existing methods in knowledge extraction can identify the system's strengths and weaknesses. Deeper qualitative evaluation with expert participation can lead to improved triple extraction processes and increased accuracy. Furthermore, developing new, specialized evaluation metrics for knowledge extraction can help establish more precise standards.

6.7.6 Application in Other Fields

Finally, practical applications in fields such as intelligent Question Answering (QA), automated summarization, and Social Network Analysis offer new opportunities for utilizing extracted knowledge. Using extracted triples in these fields helps create smarter systems, particularly for analyzing relationships and patterns in social networks.

6.8 Chapter Conclusion

In this chapter, the outputs of the knowledge extraction pipeline were examined quantitatively and qualitatively. General statistics showed that processing 100 biographies with 2000 characters on average yielded over 9,000 triples, demonstrating high scalability. The confidence analysis indicated

that about 60% of outputs are acceptable, and nearly 35% are high-confidence. These results show that the pipeline is capable of translating a significant portion of textual relationships into valid triples. However, the concentration of data in the mid-confidence range suggests that improvements in frame extraction and entity linking could further enhance quality. Case studies of major authors like Tolstoy, Pushkin, Dostoevsky, Márquez, and Shakespeare proved that the generated triples accurately represent vital relationships like birth, death, works, beliefs, and awards. Additionally, knowledge graph visualizations clearly showed that the final structure provides a rich network of semantic connections in a queryable form, enabling integration into Graph RAG frameworks. Overall, the results of this chapter highlight the system's key strengths: operational capacity for long texts, high accuracy in identifying key relationships, human-like evaluation with an LLM-Judge, and the ability to generate standard knowledge graphs. Simultaneously, areas such as improving frame accuracy and increasing output confidence were identified as future development paths.

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